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Manufacturing Process Optimisation and Predictive Analytics:

Leveraging Data to Minimise Giveaway in Cheese Production

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AI and Cheese: A Surprising Synergy

Knowledge Transfer Partnership (KTP) funded by Innovate UK and Butlers Farmhouse Cheeses. Umbrella project aims to develop novel business analytics and AI to amplify cheese yields and reduce giveaways.

This work involves process optimisation with Predictive Analytics and Machine Learning to save >£200K pa. It presents a unique approach for investigating product inconsistencies in manufacturing, especially in low data cases.

The Project Team



Dr Jiabin Luo
Academic Lead
Aston University



Dr Ozren Despić
Academic Supervisor
Aston University



Giwa Iziomo
KTP Associate
Aston University
(Seconded to Butlers)



Dr Viktor Pekar
Academic Support
Aston University



Dr Eliseo Perdomo
Academic Support
Aston University



Tim Fisher
Head Dairy Technologist
Butlers Cheeses

About Giwa - Presenter

- Data Scientist, leading execution of the KTP project. Interfaces between industry and academia
- 10+ years of experience applying analytics to drive value in manufacturing
- Background in Mechanical Engineering & Masters in Business Analytics
- Assuming the role of Head of Automation and AI at Butlers from August 2024

Business Problem – It starts with a question

We are making bigger cheese with high giveaways.

How can we better target weights close to the required 150g?

Business Impact

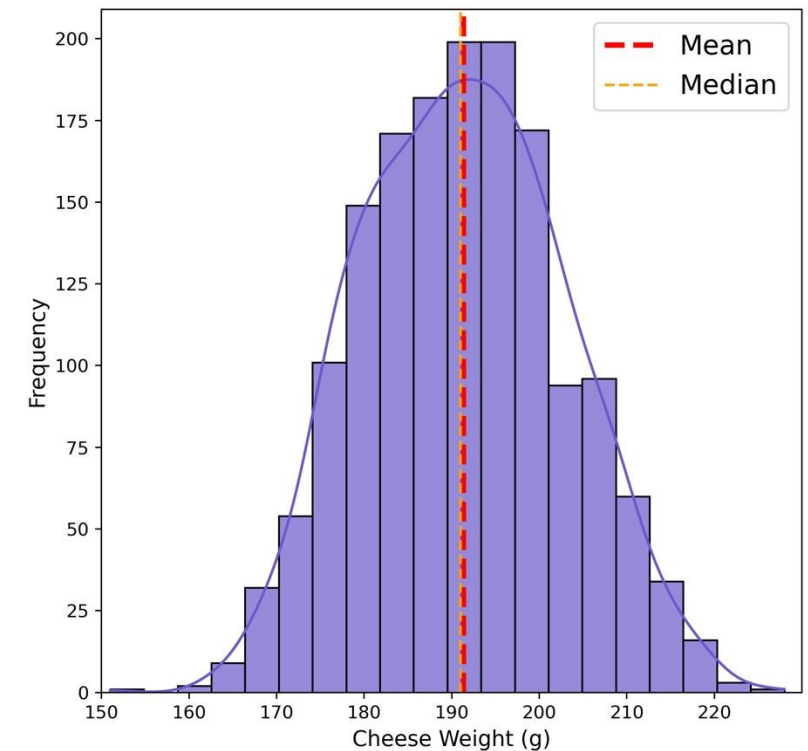
Required Weights
150g

Prod. Weights (Avg.)
180g

Giveaways
>20%

Cost per year
>£350K

Problem



Project Approach – Where is the data?

Identifying Features



Working with process experts, we developed an understanding of the problem and identified linked variables

Data Challenge



Like most Data Science projects, data on relevant variables were unavailable. So we had to create them!

Experiments Design

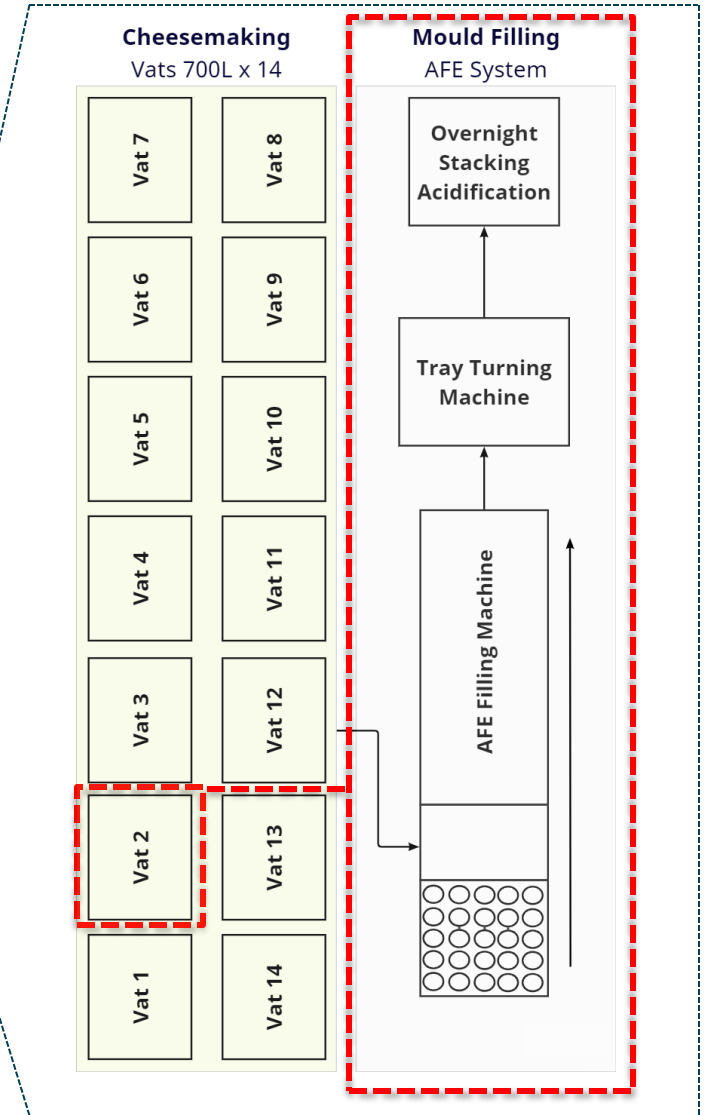


Three (3) experiments were conducted, creating varied equivalent classes and keeping other variables constant

Data Collection



Data on filling techniques, position vectors and resulting weights were collected for each cheese. 1,575 rows x 8 cols



Methods

- **Descriptive:** Exploratory Analysis, Statistical plots, Correlation heatmaps
- **Hypothesis Tests:** Independent t-test, Mann-Whitney U test, Chi-Square Statistic
- **Predictive:** Linear Regression (baseline), Random Forest Tree, XGBoost, Causal Inference

Δ in curd texture

51%
Inter-tray

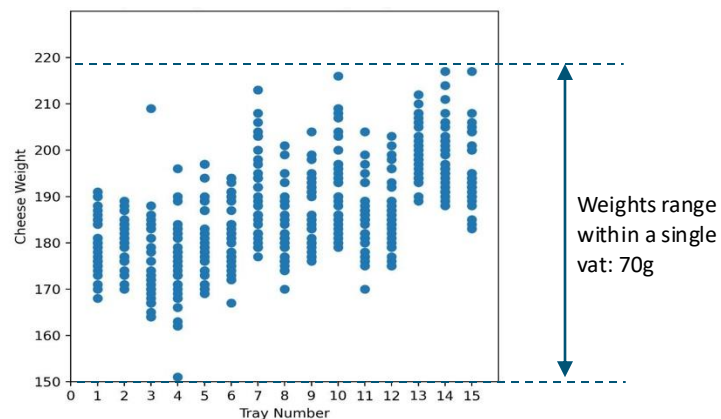
Tray design

25%
Intra-tray

Tray position

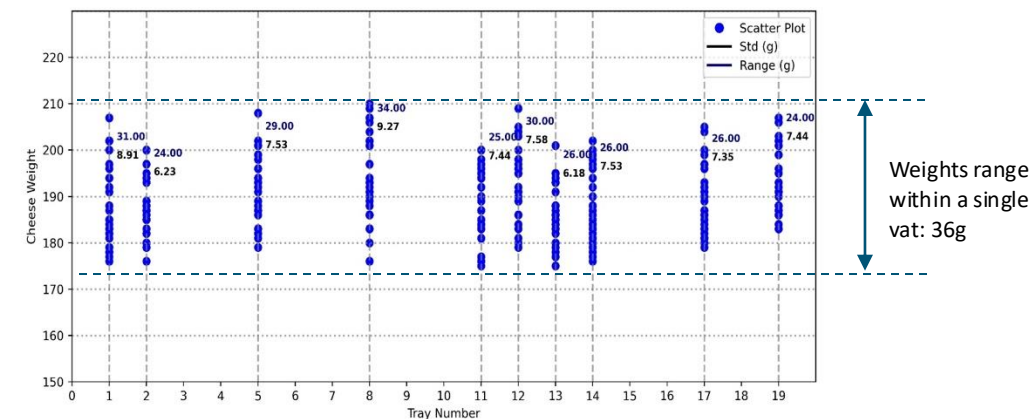
5%
Inter-tray

Fixed curd volume



- Fixed volume of curds placed into all moulds (start to finish)
- Weights between 150 – 220 grams, a **variation of 70 grams**
- Effect linked to changes in curd texture due to moisture loss over time

Variable curd volumes (trial)



- Volume of curds placed in moulds is varied throughout the vat
- Weights between 173 – 210 grams, a **variation of 36 grams**
- A software operated solution is implemented to automate variation

Recommendations & Implementations

New Mould Trays

New mould trays with better ventilation now implemented, also improving quality

New Acidification Room

A specifically designed room with controlled air conditioning to improve consistency of temperature and humidity

Standardised Mould Filling

Mould filling is a manual process. A standardised system improves consistency across operators

Automated Volume Control

Working with software engineers, a parameterised solution based on machine learning models help to vary curd volumes

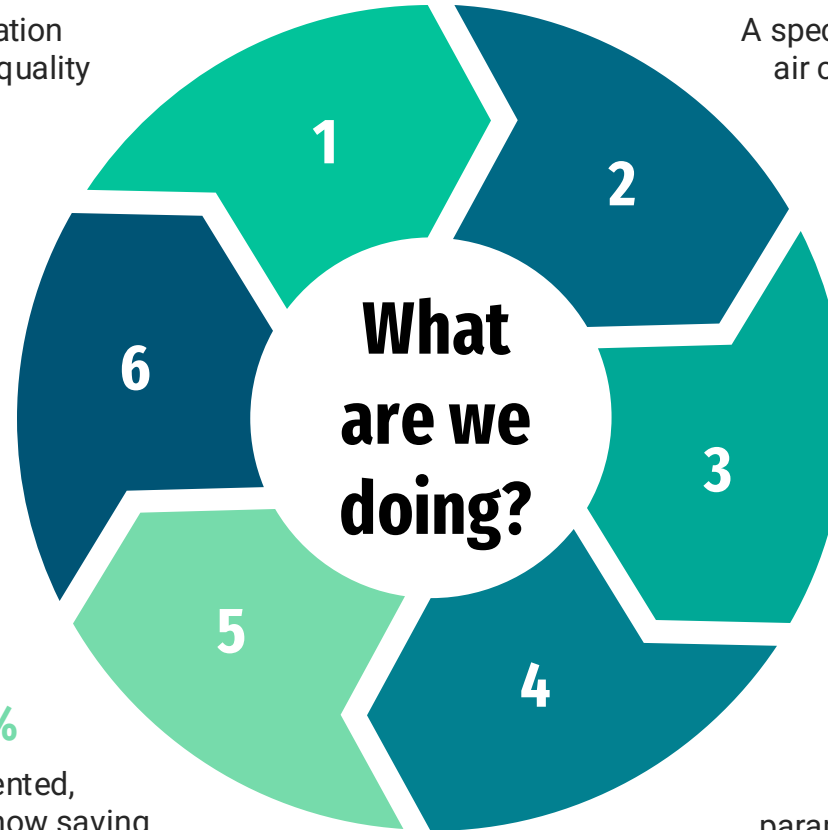
**What
are we
doing?**

Continuous Improvement

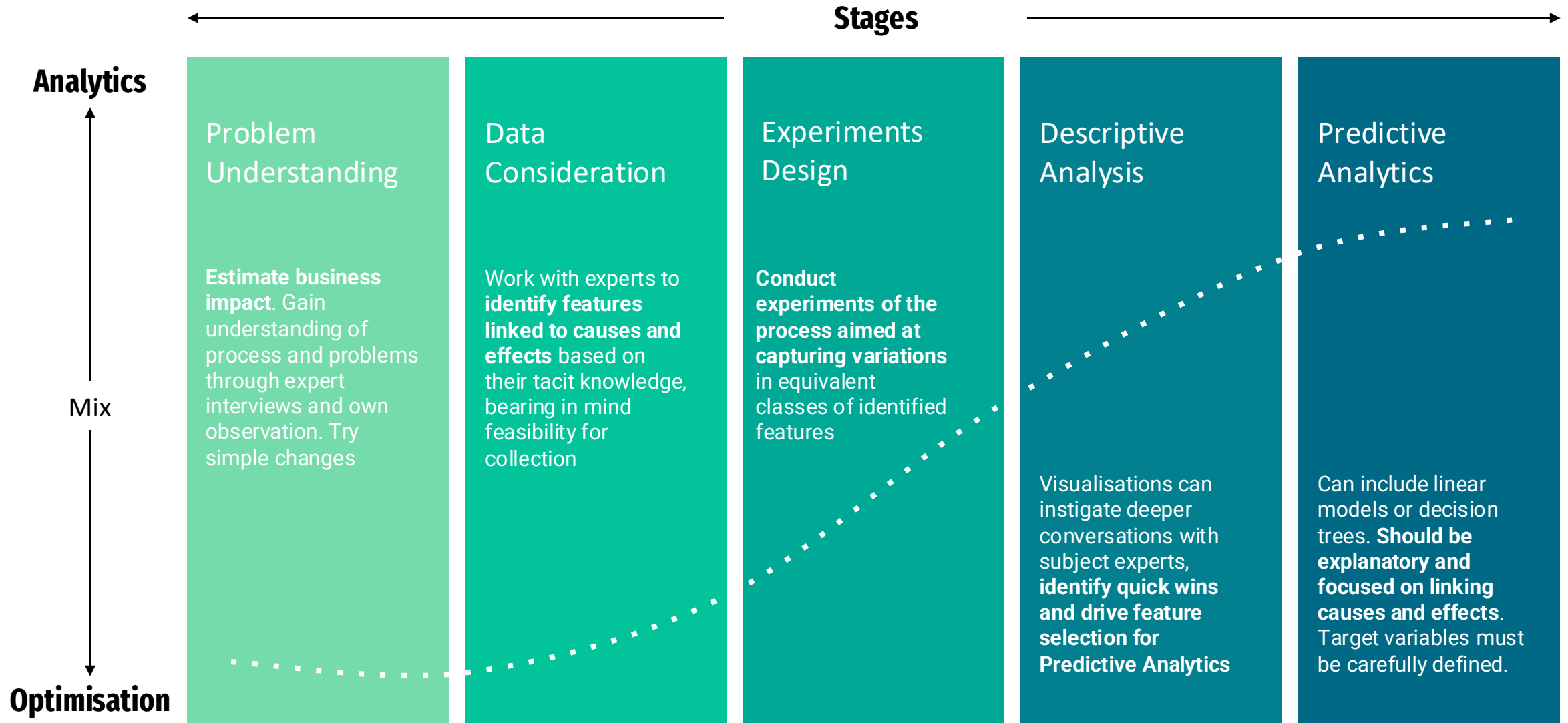
Different measures such as BI dashboards are being put in place for weight monitoring and agile improvement

Giveaway Reduced to 9%

Through the measures implemented, giveaway continues to reduce, now saving £200K per year



Summary – Bringing it all together



Questions?

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